Raghuram Kalyanam

Senior Data Scientist/Consultant

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Senior Data Scientist / AI Engineer with 14 years of professional experience, including 11 in machine learning and artificial intelligence (AI). Focused on production Large Language Model (LLM) systems: Retrieval-Augmented Generation (RAG) architecture and retrieval quality, evaluation frameworks and LLM-as-judge pipelines, prompt architecture, and AI security and guardrails. Experience taking Generative AI (GenAI) systems from prototype into monitored production in regulated and public-sector environments, including sovereign, fully self-hosted and on-premise deployments. Earlier background in time-series forecasting, reinforcement learning and MLOps across finance, automotive, aviation and construction.

Work Experience

October 2025 – Present

Senior Data Scientist / Consultant, ELCA Informatique SA, Zurich

- Developed an internal multilingual staff assistant over a corpus of work-instruction manuals for a leading Swiss collection agency, answering in German, French and Italian, grounded strictly in the source manuals, using Chainlit, OpenSearch, LiteLLM and Qwen3.
- Replaced an LLM-driven catalogue-browsing design with a dedicated search engine, cutting both latency and cost per question by a large margin, and corrected German compound-word handling in query construction to make source selection substantially more reliable.
- Designed and tuned lexical retrieval with per-language analysers, German compound-aware query shaping and LLM-generated section tags, and multimodal retrieval surfacing relevant screenshots without a vision model.
- Developed a sovereign, fully self-hosted GenAI assistant platform for the Canton of St. Gallen handling documents classified confidential and secret, with purpose-specific assistants for writing, research and summarisation, using Open WebUI, Bifrost, vLLM and Gemma.
- Designed prompt architecture for rule-following assistants, compressing large official rule sets into stable, cacheable system prompts, and evaluated three alternative designs before adopting that approach on measured evidence.
- Built evaluation frameworks with DeepEval and MLflow, combining deterministic rule metrics, cross-family LLM judges and retrieval-versus-generation failure attribution against the running system rather than a simplified stand-in.
- Mapped Data Protection Impact Assessment (DPIA) safeguards and the OWASP Top 10 for LLM Applications onto specific implemented controls, and remediated defects found in a deployment audit.
- Productionised the platform layer: zero-downtime reindexing, Keycloak OIDC single sign-on with directory-group RBAC, Prometheus and Grafana observability with per-use-case cost attribution, and GitLab CI/CD.
- Evaluated cross-encoder reranking and vector search, and documented the decision to retain lexical retrieval for a curated German corpus.

LLMs · RAG · OpenSearch · BM25 · Open WebUI · LiteLLM · Bifrost · Ollama · vLLM · Qwen3 · Gemma · DeepEval · MLflow · Chainlit · FastAPI · Keycloak · Prometheus · Grafana · Docker · GitLab CI/CD · Azure OpenAI · Python · Prompt Engineering · LLM-as-Judge · Guardrails · DPIA

July 2022 – September 2025

Senior Data Scientist, Mercedes-Benz Mobility AG, Stuttgart

- Built an LLM-powered semantic query system letting collections staff search for and identify high-risk customers in natural language, using Model Context Protocol (MCP) to translate intent into weighted SQL and vector queries across risk history, payment profiles and behavioural data, surfacing the matching customers directly.
- Built a separate semantic fraud-reasoning system for Mercedes-Benz Bank to surface abnormal credit applications, combining hybrid search over analyst comments with structured financial metrics, and using FAISS to cluster historical fraud patterns.

LLMs · LangChain · MCP · Vector Databases · ChromaDB · pgvector · FAISS · Superlinked · Ollama · LiteLLM · OpenAI API · Hugging Face · FastAPI · Docker · PySpark · SQL

January 2020 – May 2022 | **Data Scientist / Consultant, Lufthansa Industry Solutions GmbH, Raunheim**

- Developed an AI forecasting platform with rolling predictions, automated retraining and Bayesian hyperparameter optimisation, plus the ETL workflows enriching data with external regressors.
- Modelled resource and scheduling problems as reinforcement learning tasks solved with DQN variants; containerised training and inference on Docker and Kubernetes for production.
- Consulted aviation, automotive and public-sector clients; mentored junior data scientists and advised on ML architecture.

Python | PyTorch | TensorFlow | scikit-learn | Prophet | skforecast | MLflow | FastAPI | Docker | Kubernetes | Azure | Terraform | PySpark | SQL

May 2015 – October 2019 | **Research Associate, TU Kaiserslautern | LivingLab SmartOfficeSpace, Kaiserslautern**

- Developed reinforcement learning models controlling adaptive glass transparency to optimise visual and thermal comfort in real office environments, with a purpose-built daylight simulation environment generating synthetic training data and large-scale tuning via Ray Tune.
- Collaborated with SBRC (University of Wollongong, Australia) on a co-simulation framework for building performance and user comfort; six peer-reviewed publications.

Python | PyTorch | TensorFlow | Keras | RLlib | Ray Tune | OpenAI Gym | Flask | Spark | Hadoop | SQLite

November 2012 – February 2015 | **Software Engineer, Ericsson R&D Center, Chennai, India**

- Implemented VoLTE (Java) and Diameter (C++) protocols for telecom billing systems in an agile team; node performance improvements, on-call support for international customers, and release management.

Java | C++ | Design Patterns | Postgres | Wireshark | Jenkins | Jira | Git

</> Skills

Programming Languages Python, SQL, Java, C++.

GenAI & LLM Generative AI (GenAI), Large Language Models (LLMs), LangChain, Model Context Protocol (MCP), function calling and tool use, OpenAI API, Azure OpenAI Service, Azure AI Foundry, LiteLLM, Bifrost, Ollama, vLLM, Open WebUI, Chainlit, Qwen3, Gemma, Mistral, LLaMA, Hugging Face Transformers.

RAG & Retrieval Retrieval-Augmented Generation (RAG) architecture, semantic search, vector search, hybrid search, lexical search, OpenSearch, BM25 and per-language analysers, query expansion and rewriting, reranking, chunking strategy, multimodal retrieval, embedding models, vector databases (ChromaDB, pgvector, FAISS, Superlinked), Docling, Scrapy, BeautifulSoup.

Evaluation & Quality LLM evaluation, LLM-as-judge, DeepEval, GEval, golden datasets, deterministic rule metrics, benchmarking, retrieval failure attribution, regression testing, A/B experiments, MLflow experiment tracking, prompt architecture, context engineering, system prompts, token and cost optimisation, latency optimisation.

AI Security & Governance Guardrails, prompt-injection defence, adversarial test corpora, groundedness and hallucination controls, citation integrity, Responsible AI, AI governance, OWASP Top 10 for LLM Applications, Data Protection Impact Assessment (DPIA / DSFA), GDPR, data residency, sovereign AI, on-premise and self-hosted LLM deployment, Keycloak OIDC, single sign-on (SSO), RBAC.

ML & Optimisation Machine learning, deep learning, Natural Language Processing (NLP), time-series forecasting, anomaly and fraud detection, reinforcement learning, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CatBoost, RLlib, Prophet, skforecast, Ray Tune, HyperOpt, Bayesian optimisation.

Cloud & MLOps Azure, Docker, Docker Compose, Kubernetes, Helm, Terraform, Infrastructure-as-Code, microservices, GitLab CI/CD, MLOps, observability and model monitoring, Prometheus, Grafana, FastAPI, Flask, Git.

Data & Visualisation PySpark, Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit.

Ways of Working Solution architecture, technical consulting, requirements engineering, Architecture Decision Records (ADRs), stakeholder management, cross-functional collaboration, Agile and Scrum, DevOps, code review, mentoring, technical documentation.

Education

2007 - 2012 **M.Sc Mathematics and Computing**, IIT Dhanbad, India.

Languages

English	Fluent
German	C1

Achievements

- Six peer-reviewed publications in reinforcement learning and building-performance modelling (see below).
- Ranked in the top 1% of 400,000 candidates in the IIT-JEE entrance examination.
- Held a merit scholarship throughout my studies; recognised as "Rockstar" at Ericsson for performance.

Publications

- Kalyanam, Raghuram, and Sabine Hoffmann. 2021. ”A Reinforcement Learning- Based Approach to Automate the Electrochromic Glass and to Enhance the Visual Comfort” *Applied Sciences* 11, no. 15: 6949.
- Kalyanam, Raghuram, and Sabine Hoffmann. ”Reinforcement learning based approach to automate the external shading system and enhance the occupant comfort.” *AI in AEC conference*, Helsinki, Finland, March 2021.
- Kalyanam, Raghuram, and Sabine Hoffmann. ”A Novel Approach to Enhance the Generalization Capability of the Hourly Solar Diffuse Horizontal Irradiance Models on Diverse Climates.” *Energies* 13.18 (2020): 4868.
- Kalyanam, Raghuram and Sabine Hoffmann. 2016. ”Visual and thermal comfort with electrochromic glass using an adaptive control strategy.” Paper presented at the *15th International Radiance Workshop (IRW)*, Padua, Italy, August 2016.
- Boudier, Katharina, Massimo Fiorentini, Sabine Hoffmann, Raghuram Kalyanam, and Georgios Kokogiannakis. 2016. ”Coupling a thermal comfort model with building simulation for user comfort and energy efficiency.” *In Proceedings of the Central European Symposium on Building Physics (CESBP) and BauSIM*, Dresden, Germany, September 2016, 481–487.
- Hoffmann, Sabine, Andrew McNeil, Eleanor S. Lee, and Raghuram Kalyanam. 2015. ”Discomfort glare with complex fenestration systems and the impact on energy use when using daylighting control.” *In Proceedings of the Advanced Building Skins Conference*, Bern, Switzerland, November 2015.